EIA Gasoline-Demand and Inventory Surprises and Energy Futures

Research summary · Amit Aviv, Columbia University · WTI (CL) and RBOB (RB) futures, 1996–2024

Research question

Do the unexpected parts of the EIA's weekly gasoline-demand and inventory report move NYMEX crude and gasoline futures on release day (Q1), and do they predict returns *after* the release, when a trader can act (Q2)?

Data and methods

- **Point-in-time data:** release times documented for 791 of 793 reports; first-published values from archived editions (found a 30.6-million-barrel 2016 crude restatement); contract-aware dollar P&L with exchange-documented expiries.
- **Surprises:** actual minus a seasonal + AR(1) forecast, selected by out-of-sample error (44% lower demand-forecast RMSE than EIA's five-year benchmark).
- **Inference:** Newey–West regressions, Holm correction across 16 pre-registered tests; walk-forward forecasts (OOS R^2, Clark–West); stationary-bootstrap Sharpe CI, Deflated Sharpe Ratio, Hansen SPA, permutation nulls.
- **Protocol:** development 1996–2016; locked test 2017-01 to 2024-04-05, run once after a git-tagged specification freeze enforced in code.

Key findings

- **1. A real release-day effect.** Gasoline-inventory surprises move RBOB: −0.74% per sd, t = −3.67, Holm p = 0.004 out of sample, replicating development. The development crude → WTI effect (t = −4.68) did not replicate (t = −1.51).
- **2. No post-release drift.** All 8 drift coefficients Holm p = 1.00; forecast OOS R^2 −1.5% (CL) and −1.8% (RB) versus the historical mean.
- **3. The strategy failed out of sample.** Net Sharpe −0.50 against +0.35 in development; every robustness variant negative; costs were not the cause.

Table 1. Strategy performance, base costs, $1 million base.

Locked test 2017–2024	Gross Sharpe	Net Sharpe	Return / yr	Vol / yr	Max DD
Rule A (primary)	**−0.48**	**−0.50**	**−4.2%**	**8.4%**	**−34.3%**
Rule B	−0.57	−0.62	−11.7%	18.9%	−93.6%
Always long	0.13	0.07	1.5%	20.9%	−59.6%
Price-only	−0.32	−0.33	−2.4%	7.3%	−22.6%

Table 2. Main statistical results for the trading hypothesis.

Primary strategy statistic	Value	Criterion
Net Sharpe, 95% bootstrap CI	[−1.14, +0.14]	above 0 — not met
Deflated Sharpe Ratio (N = 20)	0.0008	> 0.95 — not met
Hansen SPA p-value	1.00	< 0.05 — not met

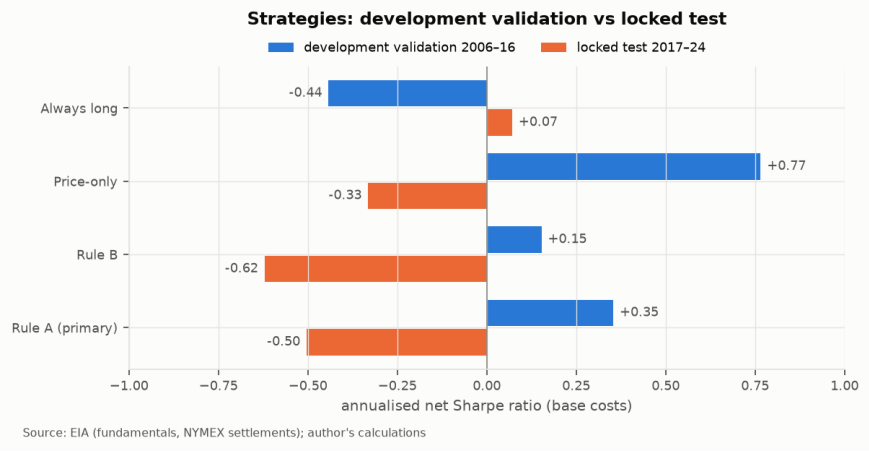


Figure 1. Net Sharpe ratios: development validation versus the locked test.

Limitations

Daily settlements miss the first minutes after the release; 379 test weeks detect only large effects; model-based surprises (no consensus or API data) attenuate coefficients; pre-2011 data uses restated history; one disclosed development amendment. Full paper: *reports/Energy_Futures_Research.pdf*.